

Definition

For all $x > 0$, we define

$$\boxed{\int_1^x \frac{1}{t} dt = \ln x} **$$

$$\int_1^x t^2 dt = \left(\frac{t^3}{3} \right) \Big|_1^x = \frac{x^3}{3} - \frac{1}{3}$$

- ✓ The expression $\ln x$ is called the natural logarithm of x .
- ✓ x has to be positive since $\int_1^x \frac{1}{t} dt$ does not exist // when $x \leq 0$.

✓ The FTC implies that

$$\frac{d}{dx} \left(\int_1^x \frac{1}{t} dt \right) = \frac{d(\ln x)}{dx}$$

$$\boxed{\frac{1}{x} = \frac{d(\ln x)}{dx}} *$$

$$\begin{aligned} \frac{d}{dx} (\log_a x) &= \frac{1}{x \ln a} \\ \text{(let } a=e. &\Downarrow \\ &\frac{d(\ln x)}{dx} = \frac{1}{x} \end{aligned}$$

✓ We can deduce the following from **

① $\boxed{\ln 1 = 0}$

why?

$$\ln 1 = \int_1^1 \frac{1}{t} dt = 0$$

$$\log_e p = \ln p.$$

$$\log_a p^r = r \log_a p$$

② $\boxed{\ln e = 1}$

why?

$$\ln e^x = \int_{e^x}^{e^x} \frac{1}{t} dt$$

$$\ln e = \log_e e = 1$$

$$x \ln e = \int_1^{e^x} \frac{1}{t} dt = x$$

$$x \ln e = x \iff \boxed{\ln e = 1}$$

$$y = \log_a p \iff p = a^y$$

Theorem

To every real number x , there's a unique (corresponding) positive real number y s.t.

$$\begin{array}{c} x = \ln y \\ \updownarrow \\ y = e^x \end{array}$$

Recall.

The function g is an inverse of f iff

$$\begin{array}{l} g(f(x)) = x \quad \text{and} \\ f(g(x)) = x \end{array}$$

✓ We say that g and f are inverses of each other.

Definition

The natural exponential fn, denoted by $\exp$ is defined by $e^x = y \iff \ln y = x \quad \forall x$, where $y > 0$

$$e^{\ln y} = y \iff \ln y = x$$

✓ We finally notice that $\ln$ and $\exp$ are inverses of each other.